Card Analytics SQL Interview Practice

Questions 1–25 • PostgreSQL • Reviewed & Adjusted Queries

BASIC

1. How many cards are in the portfolio?

```
SELECT COUNT(*) AS total_cards
FROM cards_data;
```

2. How many unique customers exist?

```
SELECT COUNT(DISTINCT customer_id) AS unique_customers
FROM cards_data;
```

3. What is the average credit limit?

```
SELECT '$' || ROUND(AVG(credit_limit), 0) AS avg_credit_limit
FROM cards_data;
```

4. Which card type has the most cards?

```
SELECT card_type, COUNT(*) AS total_cards
FROM cards_data
GROUP BY card_type
ORDER BY total_cards DESC
LIMIT 1;
```

5. Which network has the most cards?

```
SELECT card_network, COUNT(*) AS total_cards
FROM cards_data
GROUP BY card_network
ORDER BY total_cards DESC
LIMIT 1;
```

INTERMEDIATE

6. Which card type has the highest credit exposure?

```
SELECT card_type, SUM(credit_limit) AS card_exposure
FROM cards_data
GROUP BY card_type
ORDER BY card_exposure DESC
LIMIT 1;
```

7. What percentage of cards are active?

```
SELECT COUNT(*) AS total_cards,
       COUNT(CASE WHEN LOWER(card_status) = 'active' THEN 1 END) AS active_cards,
       ROUND(COUNT(CASE WHEN LOWER(card_status) = 'active' THEN 1 END) * 100.0 / COUNT(*), 2) AS active_card_percentage
FROM cards_data;
```

8. What is the contactless adoption rate?

```
SELECT COUNT(*) AS total_cards,
       COUNT(CASE WHEN LOWER(contactless) = 'yes' THEN 1 END) AS contactless_cards,
       ROUND(COUNT(CASE WHEN LOWER(contactless) = 'yes' THEN 1 END) * 100.0 / COUNT(*), 2) AS contactless_adoption_rate
FROM cards_data;
```

9. What is the virtual-card adoption rate?

```
SELECT COUNT(*) AS total_cards,
       COUNT(CASE WHEN LOWER(card_mode) = 'virtual' THEN 1 END) AS virtual_cards,
       ROUND(COUNT(CASE WHEN LOWER(card_mode) = 'virtual' THEN 1 END) * 100.0 / COUNT(*), 2) AS virtual_card_adoption_rate
FROM cards_data;
```

10. Which network has the highest average credit limit?

```
SELECT card_network, ROUND(AVG(credit_limit), 2) AS avg_credit_limit
FROM cards_data
GROUP BY card_network
ORDER BY avg_credit_limit DESC
LIMIT 1;
```

11. Which card type has the highest inactive rate?

```
SELECT card_type, COUNT(*) AS total_cards,
       COUNT(CASE WHEN LOWER(card_status) <> 'active' THEN 1 END) AS inactive_cards,
       ROUND(COUNT(CASE WHEN LOWER(card_status) <> 'active' THEN 1 END) * 100.0 / COUNT(*), 2) AS inactive_card_percentage
FROM cards_data
GROUP BY card_type
ORDER BY inactive_card_percentage DESC
LIMIT 1;
```

12. How many customers have multiple cards?

```
SELECT cards_count, COUNT(*) AS customers
FROM (
    SELECT customer_id, COUNT(card_id) AS cards_count
```

```
    FROM cards_data GROUP BY customer_id
) AS customer_cards
GROUP BY cards_count
ORDER BY cards_count DESC;

-- Top 10 customers
SELECT customer_id, COUNT(card_id) AS cards_count
FROM cards_data
GROUP BY customer_id
ORDER BY cards_count DESC, customer_id
LIMIT 10;
```

ADVANCED

13. Who are the top 10 customers by total credit exposure?

```
SELECT customer_id, SUM(credit_limit) AS total_credit_exposure
FROM cards_data
GROUP BY customer_id
ORDER BY total_credit_exposure DESC
LIMIT 10;
```

14. What percentage of exposure comes from the top 1% of customers?

```
WITH customer_exposure AS (
    SELECT customer_id, SUM(credit_limit) AS total_exposure
    FROM cards_data GROUP BY customer_id
),
top_1_percent AS (
    SELECT * FROM customer_exposure
    ORDER BY total_exposure DESC
    LIMIT (SELECT CEIL(COUNT(*) * 0.01)::INT FROM customer_exposure)
)
SELECT ROUND(SUM(total_exposure) * 100.0 /
             (SELECT SUM(total_exposure) FROM customer_exposure), 2)
       AS top_1_percent_exposure_pct
FROM top_1_percent;
```

15. Which card type contributes the most exposure per card?

```
SELECT card_type,
       SUM(credit_limit) AS total_credit_exposure,
       COUNT(card_id) AS total_cards,
       ROUND(SUM(credit_limit) / NULLIF(COUNT(card_id), 0), 2) AS exposure_per_card
FROM cards_data
GROUP BY card_type
ORDER BY exposure_per_card DESC
LIMIT 1;
```

16. How has card issuance changed year over year?

```
WITH yearly_issuance AS (
    SELECT EXTRACT(YEAR FROM issue_date)::INT AS issue_year,
           COUNT(card_id) AS cards_count
    FROM cards_data
    GROUP BY EXTRACT(YEAR FROM issue_date)
)
SELECT issue_year, cards_count,
       LAG(cards_count) OVER (ORDER BY issue_year) AS previous_year_cards,
       ROUND((cards_count - LAG(cards_count) OVER (ORDER BY issue_year)) * 100.0 /
             NULLIF(LAG(cards_count) OVER (ORDER BY issue_year), 0), 2) AS yoy_change_pct
FROM yearly_issuance
ORDER BY issue_year;
```

17. How has virtual-card adoption changed over time?

```
SELECT EXTRACT(YEAR FROM issue_date)::INT AS issue_year,
       COUNT(*) AS total_cards,
       COUNT(CASE WHEN LOWER(card_mode) = 'virtual' THEN 1 END) AS virtual_cards,
       ROUND(COUNT(CASE WHEN LOWER(card_mode) = 'virtual' THEN 1 END) * 100.0 / COUNT(*), 2) AS virtual_card_adoption_pct
FROM cards_data
GROUP BY EXTRACT(YEAR FROM issue_date)
ORDER BY issue_year;
```

18. Which card type has the highest contactless adoption?

```
SELECT card_type, COUNT(*) AS total_cards,
       COUNT(CASE WHEN LOWER(contactless) = 'yes' THEN 1 END) AS contactless_cards,
       ROUND(COUNT(CASE WHEN LOWER(contactless) = 'yes' THEN 1 END) * 100.0 / COUNT(*), 2) AS contactless_adoption_rate
FROM cards_data
GROUP BY card_type
ORDER BY contactless_adoption_rate DESC
LIMIT 1;
```

19. What is the credit exposure associated with inactive cards?

```
SELECT card_status, COUNT(*) AS card_count, SUM(credit_limit) AS credit_exposure
FROM cards_data
WHERE LOWER(card_status) <> 'active'
GROUP BY card_status
```

```
ORDER BY credit_exposure DESC;

-- Total non-active exposure
SELECT SUM(credit_limit) AS total_inactive_card_exposure
FROM cards_data
WHERE LOWER(card_status) <> 'active';
```

20. Which customers hold multiple cards?

```
SELECT customer_id, COUNT(card_id) AS total_cards
FROM cards_data
GROUP BY customer_id
HAVING COUNT(card_id) > 1
ORDER BY total_cards DESC, customer_id
LIMIT 10;
```

21. What credit exposure is expiring each year?

```
SELECT EXTRACT(YEAR FROM expiry_date)::INT AS expiry_year,
       COUNT(card_id) AS cards_expiring,
       SUM(credit_limit) AS credit_exposure
FROM cards_data
GROUP BY EXTRACT(YEAR FROM expiry_date)
ORDER BY expiry_year;
```

22. Which customers have cards expiring soon?

```
-- 'Soon' is defined here as cards expiring during 2026.
SELECT customer_id, expiry_date, COUNT(card_id) AS cards_count
FROM cards_data
WHERE EXTRACT(YEAR FROM expiry_date) = 2026
GROUP BY customer_id, expiry_date
ORDER BY expiry_date ASC
LIMIT 10;
```

23. Rank customers by total credit exposure.

```
SELECT customer_id, SUM(credit_limit) AS total_credit_exposure,
       RANK() OVER (ORDER BY SUM(credit_limit) DESC) AS exposure_rank
FROM cards_data
GROUP BY customer_id
ORDER BY exposure_rank
LIMIT 10;
```

24. Find the top card type within each network.

```
WITH network_card_types AS (
    SELECT card_network, card_type, COUNT(*) AS card_type_count
    FROM cards_data
    GROUP BY card_network, card_type
),
ranked_card_types AS (
    SELECT card_network, card_type, card_type_count,
           RANK() OVER (PARTITION BY card_network ORDER BY card_type_count DESC) AS type_rank
    FROM network_card_types
)
SELECT card_network, card_type, card_type_count
FROM ranked_card_types
WHERE type_rank = 1
ORDER BY card_network;
```

25. Find customers whose exposure is above the overall average.

```
WITH customer_exposure AS (
    SELECT customer_id, SUM(credit_limit) AS total_credit_exposure
    FROM cards_data
    GROUP BY customer_id
)
SELECT customer_id, total_credit_exposure
FROM customer_exposure
WHERE total_credit_exposure > (
    SELECT AVG(total_credit_exposure) FROM customer_exposure
)
ORDER BY total_credit_exposure DESC
LIMIT 10;
```

Key Project Notes

• The dataset does not contain a Premium card category, so Premium-card-specific analysis was not included.

• card_status contains active, blocked, expired, and lost. For Q19, non-active cards are treated as inactive exposure.

• For Q12, the analysis found 3 cards as the maximum observed per customer, with 1,253 customers holding 3 cards.

• Percentage results are kept numeric so they can be sorted and reused in further analysis.

• Q16 uses LAG() to compare each year's issuance with the previous year.

• Q24 uses RANK() with PARTITION BY card_network to rank card types separately within each network.

• Q25 compares each customer's total exposure with the average of all customer-level total exposures.